

ODE 小测

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求解下列常微分方程：

1. $x \frac{dy}{dx} + 2\sqrt{xy} = y, \quad (x < 0), \quad y|_{x=-1} = -1.$

2. $x \frac{dy}{dx} = (6 - x^2y)y.$

3. $e^y dx - x(2xy + e^y)dy = 0.$

4. $y'' = [(y')^2 + 1]y', \quad y|_{x=0} = \frac{\pi}{2}, \quad y'|_{x=0} = 1.$